



18-649 Guest Lecture: **Machine Learning**

Tianshu Huang

- (1) When should I use ML / DL?
- (2) How do I develop and deploy ML to the edge?



This lecture brought to you by:

RadarML

Interested in machine learning research and working with **real systems**, not just canned datasets?

Embedded HW/SW development for learning-enabled radar applications

Machine learning for radar and radar+X fusion

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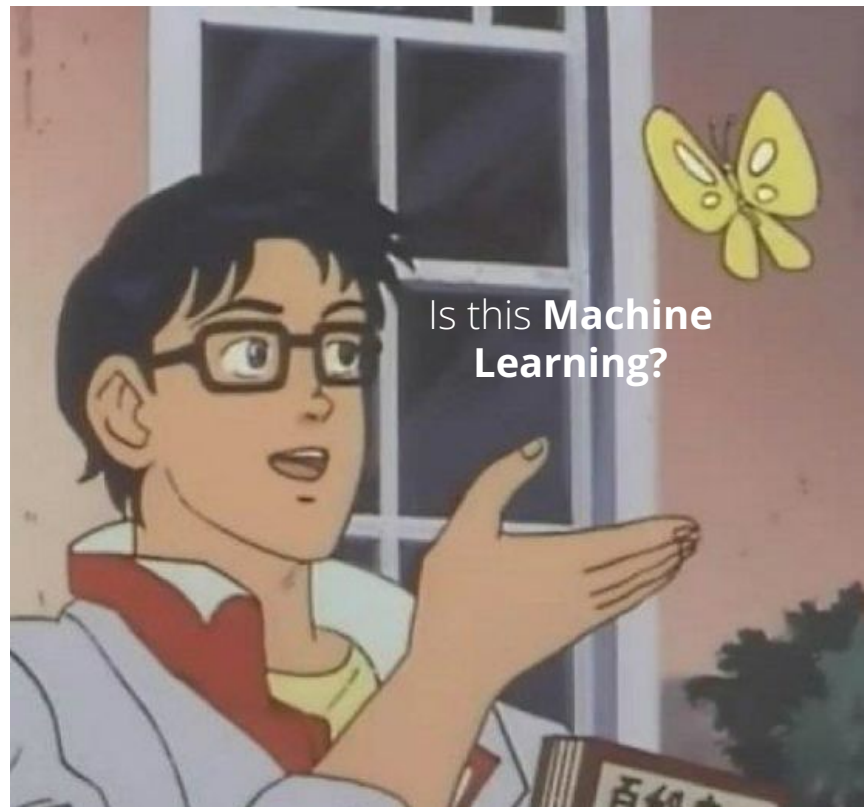
“AI”



Machine Learning

— \$\$\$

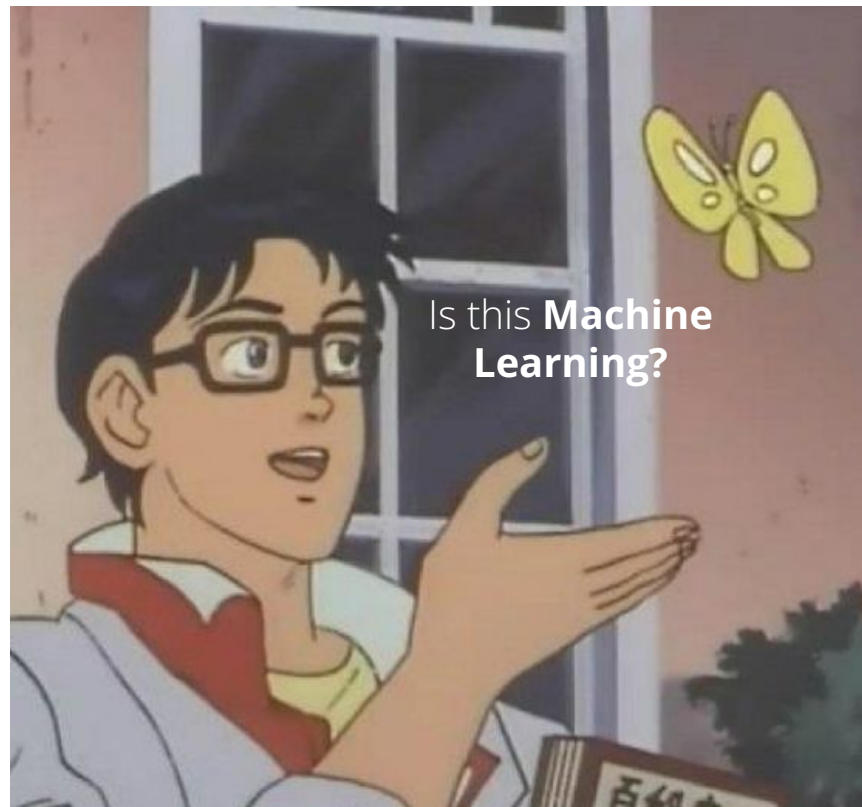
— model fitting



Machine Learning

Empirical risk minimization:

1. Data
2. Risk function (e.g. accuracy, loss)
3. What model minimizes the risk function on the data?



Machine Learning

Example: digit classification

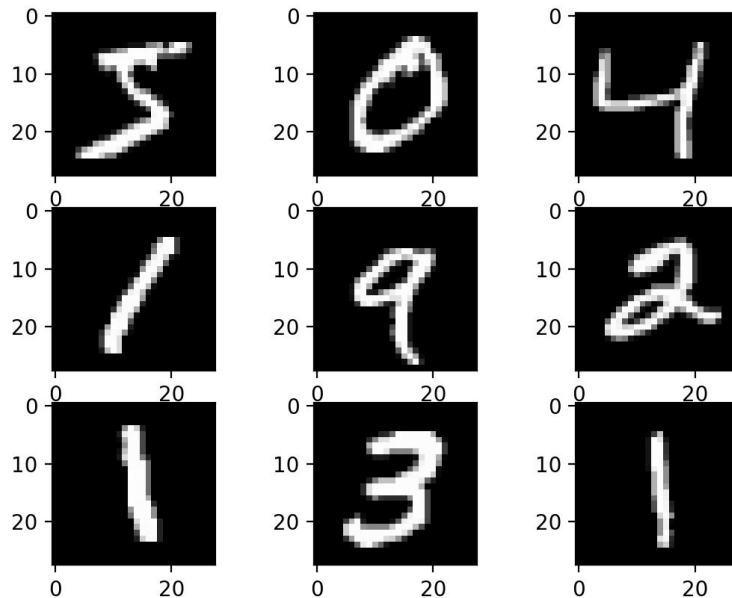
No data: Manually specified features

Some data: Compare with data

manually extract features

→ model

A lot of data → deep learning



Machine Learning

random forest

k-nearest neighbors

linear regression ; variants

decision, other unsupervised

deep learning

THIS IS YOUR MACHINE LEARNING SYSTEM?

YUP! YOU POUR THE DATA INTO THIS BIG PILE OF LINEAR ALGEBRA, THEN COLLECT THE ANSWERS ON THE OTHER SIDE.

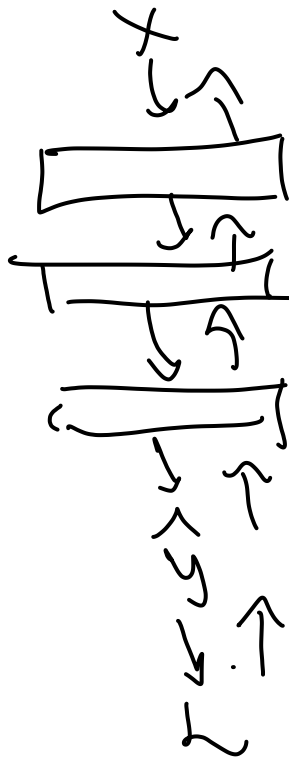
WHAT IF THE ANSWERS ARE WRONG?

JUST STIR THE PILE UNTIL THEY START LOOKING RIGHT.



<https://xkcd.com/1838/>

Deep Learning

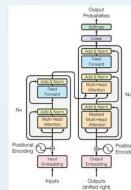


STOP DOING DEEP LEARNING

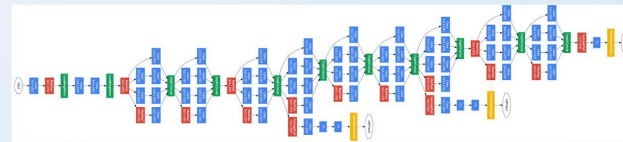
- PERCEPTRONS WERE ONLY EVER MEANT TO BE FULLY CONNECTED
- THOUSANDS OF PAPERS yet NO REAL-WORLD USE FOUND for going deeper than ONE LAYER
- Wanted to add more nonlinearity anyway for a laugh? We had a tool for that: It was called “KERNEL METHODS”
- “Yes please give me a network that can PAY ATTENTION TO ITSELF. Please give me PRETRAINED WEIGHTS for my YOLO-9000” - Statements dreamed up by the utterly Deranged

LOOK at what “Research Scientists” have been demanding your Respect for all this time, with the statistical methods & optimization algorithms we built for them

(This is REAL “Deep Learning”, done by REAL “ML Engineers”):



??????



????????????????????

“Hello I would like to learn **1.6 TRILLION** parameters please”

They have played us for absolute fools

https://www.reddit.com/r/okbuddypd/comments/n2m6vz/stop_doing_deep_learning/

Deep Learning



The Higher Mysteries

NOoO!! You have to understand Deep learning!!!

Merging Models modulo Permutation Symmetries (Ainsworth et al., 2022)

Neural Networks are Decision Trees (Aytekin, 2022)

Universal Approximation Theorem

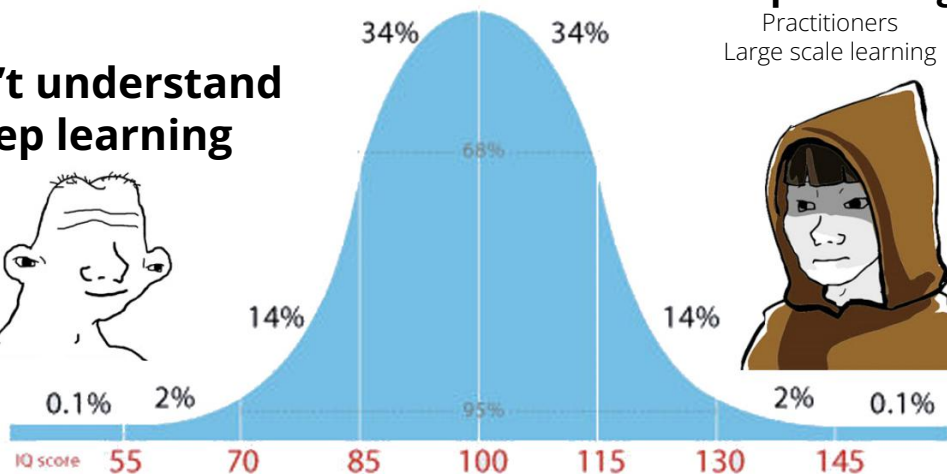
...

**I don't understand
deep learning**



**No one understands
deep learning**

Practitioners
Large scale learning



The Natural Image Manifold

The **natural image manifold** is a surface embedded in a **high dimensional space** that is:

- (1) **low-dimensional**,
- (2) **highly nonlinear**, and
- (3) **locally smooth**.

100 x 100 pt grayscale

- 10,000 dims

- random image $\rightarrow$ noise

$\Rightarrow$ natural images are low dimensional

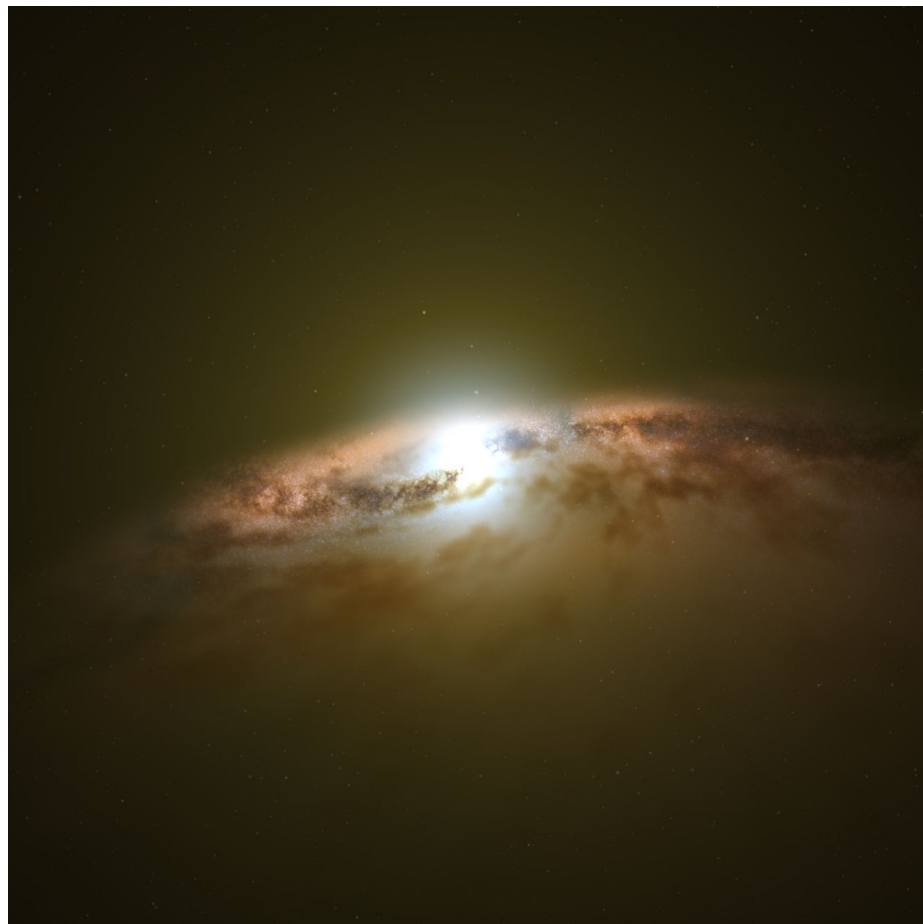
- image + image $\neq$ image $\Rightarrow$ nonlinear

- image + noise = image $\Rightarrow$ smooth

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- images, video

- text

- audio

- health data* (except continuous monitoring)

- few sensors, few observations

object detection

power analyzers → "constructed" time-series, maybe DL
→ not DL

Practical Deep Learning



See **Adversarial Robustness**.
Image from "Robust Physical-World
Attacks on Deep Learning Models,"
Eykholt et al, 2018.

Pre-Trained Models & Transfer Learning



[https://www.reddit.com/r/machinelearningmemes/comments/g1mxz8/transfer_learning_101/]

Practical Deep Learning

- Train from scratch?
- Transfer learn?
- Fully pre-trained model?



Deep Learning on the Edge



Please inspect your power cables if you own a RTX 4090, even if you don't use it for deep learning.

Deep Learning on the Edge

- Trained a neural network with a given architecture
- Weights are 32-bit float
- Fixed power budget

- offloading
- specialized HW
- Shrink the model size (quantization)
- adaptive inference

Simplest is Best

- Cloud offloading
- Reduce input resolution
- Don't use deep learning



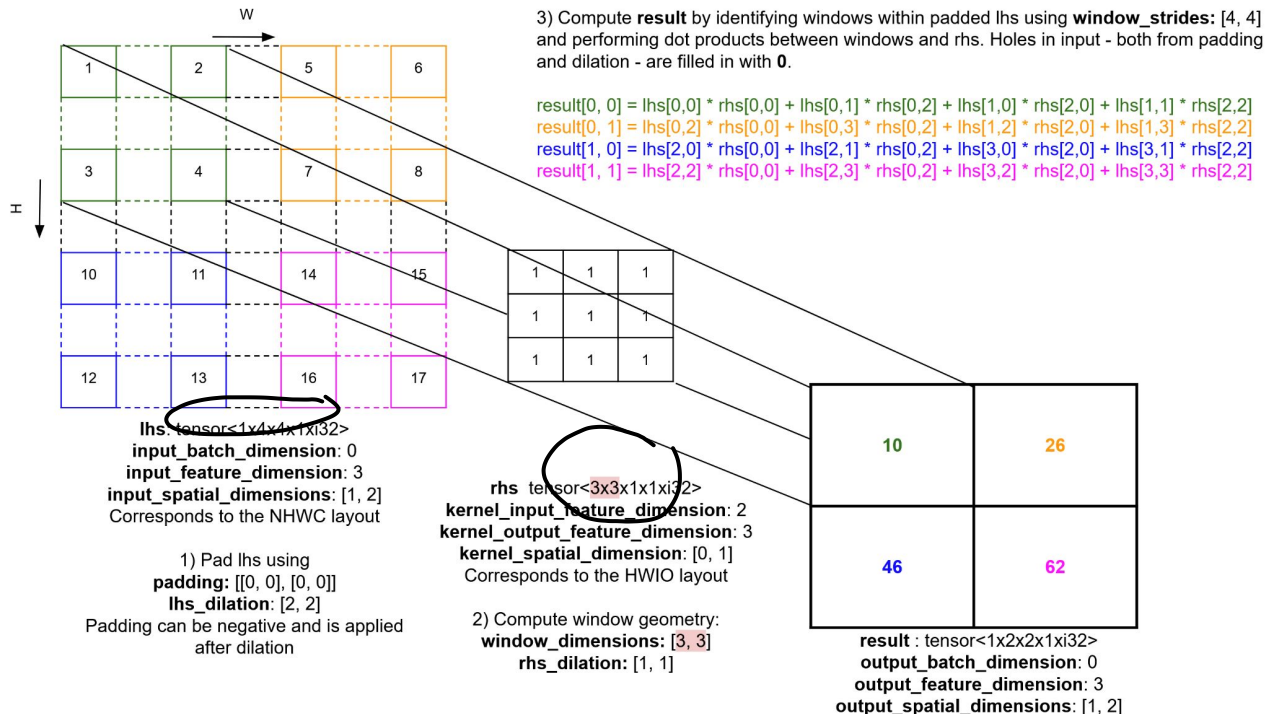
Use better software

- Compilers, e.g. XLA
- Custom CUDA / accelerator code

$$\textcircled{1} a = b + c$$

$$\textcircled{2} e = a + d$$

- ~~Arithmetic~~ stall
- cache



Quantization

flexible

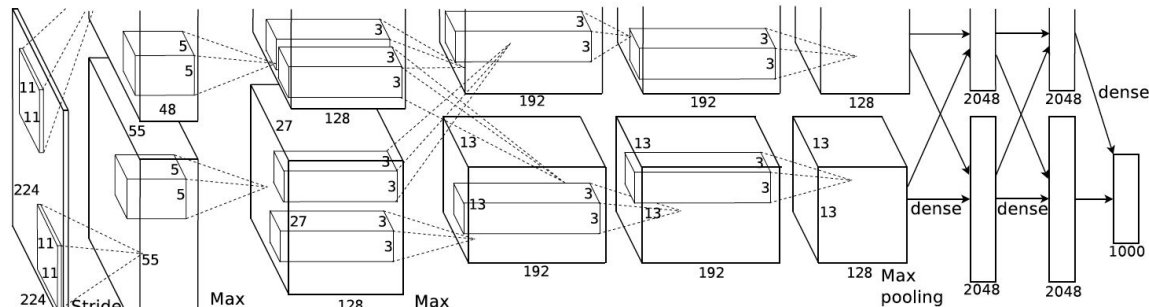
- float32 $\rightarrow$ float16
- even better: int8
- even even better: int4
- binary, ternary is even possible!

not desirable

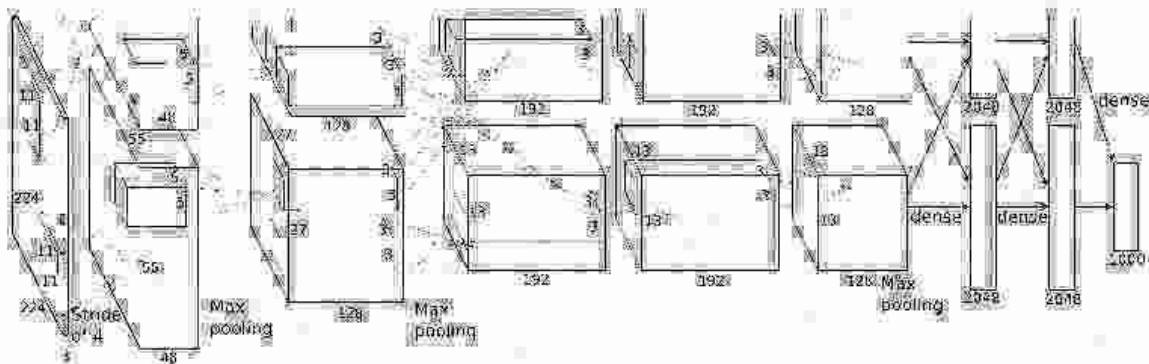
binary { 0 1
-1 1

ternary -1 0 1

"ImageNet Classification with Deep Convolutional Neural Networks," Krizhevsky et al., 2012 (AlexNet - First "true" ConvNet)

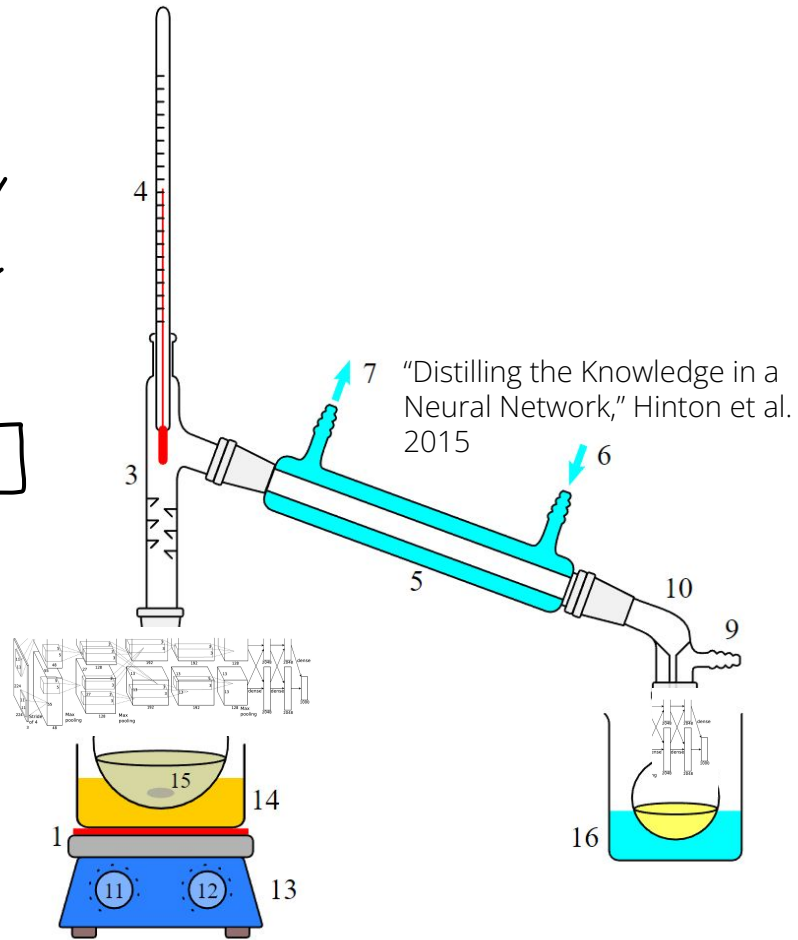
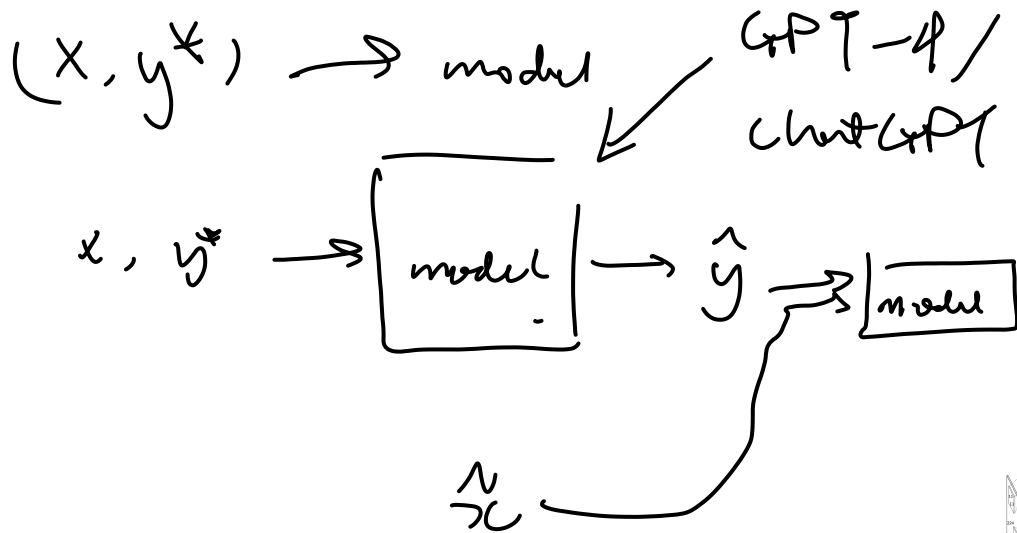


Deep Learning



Quantized Deep Learning

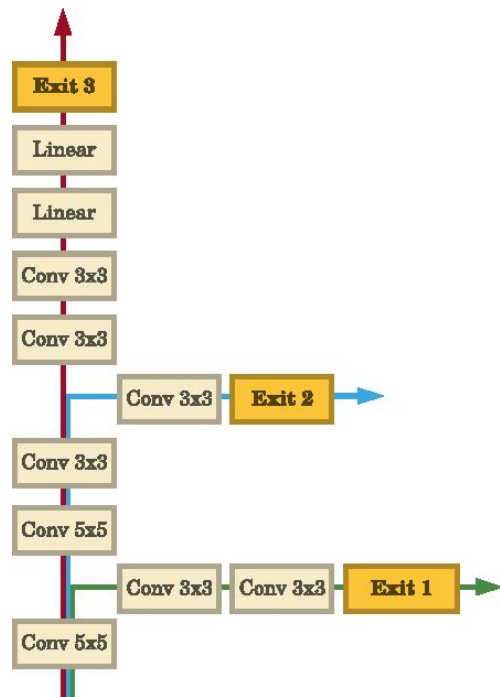
Distillation



Adaptive Inference

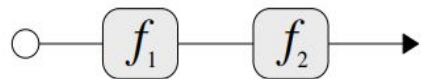
- Different numbers of layers
- Different input resolution
- ...

"BranchyNet: Fast inference via early exiting from deep neural networks,"
Teerapittayanon, 2017

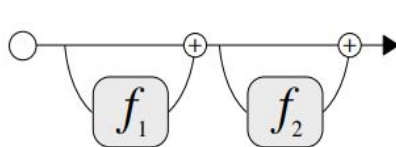


"Convolutional Networks with Adaptive Inference Graphs," Veit & Belongie, 2018

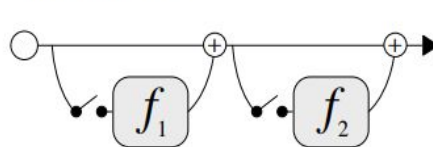
Traditional feed-forward ConvNet:



ResNet:



ConvNet-AIG:





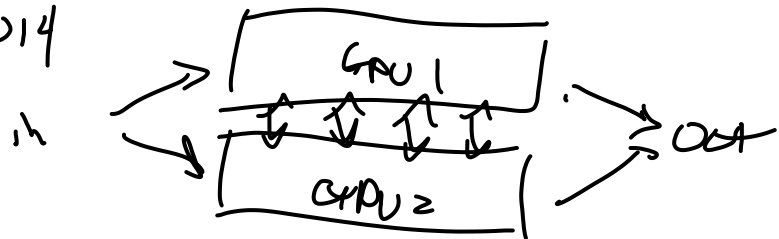
TL;DR

1. Don't use machine learning
2. Don't use deep learning
3. Don't use deep learning on the edge (do it in the cloud)
4. Don't train deep learning models (use a pre-trained model optimized for edge deployment)
5. Don't train deep learning from scratch (use transfer learning)
6. Give up (or take a machine learning class)

Q&A

- Distributed/federated training
- Edge training
- GPU vs CPU vs TPU
- Deep learning frameworks
- ML accelerators
- ML Compilers
- Efficient Architectures
- Anything ML related

Alexander, 2014



MOE

